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AI-Driven Spectrum Management: Using Machine Learning and Agentic Intelligence for Dynamic Wireless Optimization

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Abstract

Recent theoretical advances in machine learning enable a new approach for reformulation, self-learning, and specification of complex problems. We propose a novel architecture and a set of techniques that, together, can be regarded as an instantiation of SDN, knowledge-based software-defined communications, and machine learning for self-management. By combination of novel SDN controllers and novel self-learning algorithms, users change their operational parameters at runtime for profit maximization and/or resource conservation. The latter either optimally selects dynamically the communication resources for added constraints, or transmits continuously variable signals that achieve a desired target signal at the receiver even while the channels and the receiver location are unknown. All these techniques are context-aware and resort to state-of-the-art models of the channel/receiver environment but do not consult the operator's centralized controller so as to allow for scalability.

While cognitive networking has recently emerged as a highly promising paradigm for the future mobile networks, it is unclear whether global utility optimization of the network state via such techniques can be achieved. In addition, high idealization is imposed to the network elements regarding a comprehensive state knowledge which may be impractical and computationally prohibitive in reality. We exploit recent theoretical advances in distributed computation for novel datacenter management schemes that gradually localize non-redundant measurements of stochastic problems and converge to a unique global gradient. Social comparator structures are proposed to exploit this locality into the network elements while guaranteeing joint compact state-space convergence and near-optimality; as a dynamic mechanism to incentivize network elements via a new reward to unlike measurements. Feedback of the elements' utility states is a prerequisite, thus allowing extension of the approach to cognitive networks.

Keywords: AI-driven spectrum management, machine learning, agentic intelligence, dynamic wireless optimization, spectrum allocation, cognitive radio, autonomous networks, reinforcement learning, multi-agent systems, wireless communication, real-time optimization, self-organizing networks, 5G, 6G, radio resource management, intelligent orchestration, adaptive networks, edge AI, spectrum efficiency, network automation.

1. Introduction

The global growth in wireless services creates an exhaustion of available spectrum. However, it is noted that most of the spectrum is allocated to licensed users who occasionally use the spectrum, thus under-utilizing the spectrum. Therefore, it has become a necessity to devise a new mechanism to distribute the spectrum in order to avoid wasted spectrum. Recent advances in wireless technology can be used to gain knowledge of the environment and alter the transmitter parameters, thus allowing for new spectrum management. The intelligent agent is an effective knowledge representation and distributed decision-making mechanism that can be used for environment perception and exploitation. A distributed intelligent agent-based spectrum management approach is proposed to identify completely free and unused channels. The proposed approach can be used as a foundation for developing an intelligent agent-based spectrum management system in heterogeneous wireless access networks. An extended multi-agent system is modeled and implemented to demonstrate the feasibility of the proposed approach using software agents to find the completely free and unused channels.

It is well known that there is a limited amount of radio frequency spectrum, which is divided into frequency bands. Both conventional and cognitive wireless networks require a license frequency band for communication. However, it is also known that not all allocated bands are used all the time. Recent spectrum analysis across the globe has indicated that between 50 percent and 90 percent of the spectrum is under-used, depending on the band, location, time, and other parameters. Nowadays, many countries passed legislation allowing secondary users to share the primary user's licensed spectrum under certain conditions. Cognitive radio is a smart radio capable of being aware of environmental conditions and adapting its parameter according to the user's requirements. The system utilizes Wi-Fi technology with suitable transmission parameters such as transmission power, channel, modulation scheme, and coding rate to provide satisfactory quality of service. Without license accounting for the system in rural areas, the possibility of network dysfunction is less but not negligible. On the other hand, the harshness of weather conditions affects the Wi-Fi equipment's capability to self-adjust channels, which may inefficiently occupy the system. It is therefore necessary to find unused channels.

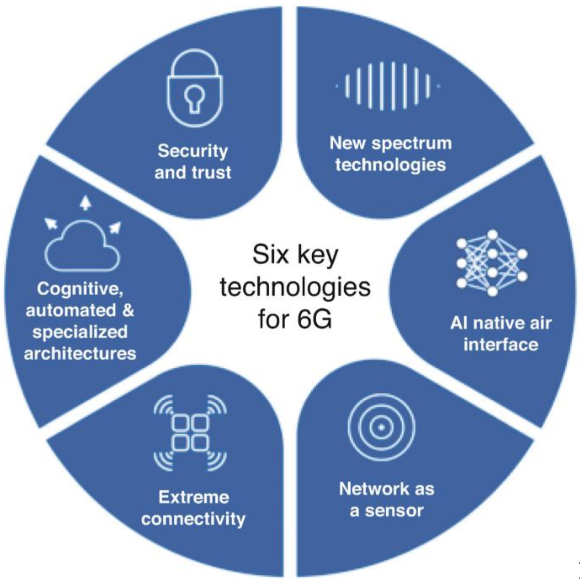

Fig 1: Artificial Intelligence for Intelligent Spectrum

2. Background on Spectrum Management

The RF spectrum is a finite natural resource [2]. The demand for its use for radiocommunication services, such as mobile telephony, digital television, internet access via satellite or microwave, and scientific measurements, is widespread and growing. The efficient management of the RF spectrum is therefore now firmly and intensely on the international agenda. The RF spectrum is a natural resource in great demand, the efficient use of which requires coordinated planning and allocation. The radiofrequency spectrum is a finite natural resource, resulting in competing demand for its use. Its economic might is illustrated by the massive revenues attracted by the recent auction of frequency bands for digital mobile telephony: these were expected to exceed 20 billion pounds Stirling. Like a natural resource, the RF spectrum is not ubiquitous. Unlike most other natural resources, the RF spectrum is non-depletable. It is a natural resource that cannot be consumed, and therefore does not run out. The RF spectrum is readily adaptable to new demands, and can be re-farmed between users and services as situations change. It is unlike the natural resources of minerals and fuels, water and air, which offer linear time exploitation profiles and do run out.

In contrast, the RF spectrum is a natural resource that can be used again and again. It is reusable. Bit rates of the order of GHz are achievable, which is an order of magnitude greater than any other transmission medium. Due to technical and economic reasons, some parts of the RF spectrum are better suited to certain uses than others, hence the concept of frequency planning. The RF spectrum has been shared over wide geographical areas by some services, for example, AM broadcasting. In some circumstances, if properly managed, spectrum can be simultaneously shared. Recognising that the RF spectrum is, in this sense, a natural resource, there is then a need for management. However, spectrum management is inherently an international activity, as regional transmitting systems must comply with the same frequency plans. The peak international spectrum management body is the International Telecommunication Union (ITU).

3. Machine Learning Fundamentals

In recent years many areas of Artificial Intelligence (AI) have seen progress in both the design of methods and in their application in actual scenarios. When AI-based applications for long-standing problems like resource management were introduced, many hoped that a new era of intelligent operation would arrive. Unfortunately, many applications do not take full advantage of AI capabilities but instead apply traditional methods with the addition of simple heuristics [3]. The wireless community experienced a change of paradigm towards distributed and autonomous resource management solutions. Centralized solutions using optimization or operation research in general can attain optimal results in terms of maximum throughput or minimum power consumption but are in nature not distributed nor real-time amenable. Solutions using heuristics like game theory, algorithms based on graph theory or a combination of both usually find equilibrium states that could be improved since they were based on wrong assumptions or applied under incorrect conditions.

A broad range of traditional non-AI methods exist to solve resource allocation, scheduling or spectrum management problems. Some methods can be categorized into binary decisions: broad classes, either as greedy or optimization. Greedy approaches, designed to myopically select the best local options, can be easily distributed but require an accurate model that invariably does not hold in reality. Optimization based solvers like dynamic programming, integer programming or linear programming can exactly find optimal solutions whenever a model is defined, but their computational complexity grows exponentially with the dimensionality. As of now, they are usually centralized and require lengthy calculations even for moderate size instances. The introduction of Machine Learning (ML), and in particular Reinforcement Learning (RL), has transformed the way many decision making problems are understood and addressed. Decision problems in which apart from the current state at hand neither the processes nor the models are known fall into the category of Markov Decision Processes (MDP), and can be solved using RL. In MDP the decision maker, the agent, interacts with the environment through the agents' actions that influence the state of the environment and return a reward.

3.1. Supervised Learning Machine learning techniques can be categorized as supervised or non-supervised. Supervised algorithms are trained with inputs X alongside their corresponding outputs Y. Using training data composed of various input-output pairs, these algorithms learn mappings from inputs to outputs. The trained algorithms are then input with test data W and make predictions of corresponding Y-values, or results. Common supervised methods include support vector machines (SVMs), neural networks, decision trees, logistic regression, among others.

$$\max_{\mathbf{A}} \sum_{i=1}^{N} U_i(f_i, P_i)$$

- $U_i(f_i, P_i)$: utility (e.g., throughput, SNR) for user i on frequency f_i with power P_i
- $\mathcal{F}$: available frequency bands
- $\mathbf{A}$: allocation vector for users

Eqn.1:Spectrum Allocation as Optimization Problem

Three different machine learning supervised approaches (algorithms) are evaluated based on their prediction accuracy over different validation ranges $x < 50$, $50 \leq x < 150$, and $x \geq 150$ where x refers to input contracting ratio. As data sets are randomly generated, each algorithm runs 10 times for each validation range and calculates mean prediction accuracies. In addition, input variables N, K, BSN, and D are evaluated for their relative importances in predicting ON nodes.

Support Vector Machines (SVMs) work by mapping a training set onto a lower-dimensional space where the training samples belonging to different categories are separated by a hyperplane. Mathematically, given training points x1, x2,⋯,xn: Xi ∈, 1 ≤ i ≤ n, the goal is to separate two classes. In addition, in order to avoid overfitting, the models are not required to perfectly classify the training points but require them to be separated with a margin. More explicitly, SVMs aimed to solve the following quadratic optimization problem: W = Minкей ∈) . With each variable Wj denoting a hyperplane's coefficient and the number of support vectors S, ∑j=1n lockingj.

The goal of this method is to classify data by deriving a linear function that provides separation between classes. The hyperplane will be maximized to choose the appropriate margin from the samples [4]. The reason for this choice is that neural networks are suitable for approximating mathematical functions and performing complex predictions in real-time.

3.2. Unsupervised Learning Given the strict requirements imposed by the wireless communication access auction, Artificial Intelligence (AI)-driven data-driven dynamic spectrum management systems are explored for distributed spectrum management by showcasing 1) unsupervised learning, and 2) reinforcement learning.

Data collection by approaching against the used fixed Node B and acquiring Angles of Arrival (AoA) measurements and Signal-to-Noise Ratio (SNR) of received wideband transmissions, the problem of spectrum hole detection and bandwidth and communications scheduling allocations by multi-class classification is framed. A data-driven framework for collaborative wideband spectrum sensing and scheduling for networked unmanned aerial vehicles (UAVs) is proposed, where UAVs intend to seek each spectrum hole in the UHF bands for terrestrial Microwave Integrated Circuits (MIC) wireless communications. The sampled wideband I/Q time-series signals are converted into time-series AoA measurements and SNR scalar values. Given extensive AoA and SNR training dataset samples, a supervised multi-class classification GBDT algorithm is developed for estimating the 3D geographic parameters of the wideband spectrum holes. For a near-real-time solution under a reasonable sub-optimality bound on the noise operator sparsity, the explainable bandit-based scheduling allocations to the prior-detected spectrum holes consumed by networked UAVs are proposed.

Thorough numerical results and physical models are given in Theorem 1, Equation (7) and (8), and they demonstrate the effectiveness of the proposed AI/ML approach with its real-world problems. The explained bandwidth allocation distributions of allocations, estimation accuracy, and data processing time show the correctness, importance, and significant computational efficiency of the proposed AI-driven systems. The reconstructed true UHF RFID and RF MIC spectrum sensing maps show the uniqueness of the spectrum holes and angles across all 3D channels. Last but not least, the individual and collective classification accuracies and efficiency enhancements illustrate the advantage of the cooperative GBDT classifiers over the simple decision tree and KNN models [5].

3.3. Reinforcement Learning Reinforcement learning (RL) refers to the process when an agent interacts with its environment, performing actions in order to obtain a reward or to learn a desired behavior. Originally inspired by theories of animal learning, RL can be applied to a various range of applications, such as game playing, robotic control, and recommendation systems. After a period of successful applications, RL has attracted increased attention from business leaders or authorities who believe it will perform revolutionary changes in the world. Unlike supervised learning techniques that are trained against a fixed dataset, RL methods rely on their own interaction with the environment to explore this previously unknown domain. A large number of off-line approaches, which can train before deployment, are applicable, such as supervised or self-supervised learning. However, they are time-consuming, cannot adapt to changes during operation, require simulating resource-demanding tasks which may be infeasible, and not well generalizable. Thus, on-line approaches are pursued.

Spectrum management in the form of primary channel selection and bandwidth allocation is key for the performance of channel bonding WLANs. Since the last few years, dedicated hardware and protocols have appeared. Some implementations rely on pre-fixed rules or heuristics. They achieve substantial performance enhancements in steady deployments. However, since they do not adapt to changes in the environment, inevitable situations may occur where agent input demand and channel conditions change. Multi-agent systems using channel bonding WLANs are particularly vulnerable, since they need to cope with all-on and all-off demands or dynamic and mesh networks. In this sense, traditional rules and heuristics become highly inefficient as they experience performance degradation, larger latencies, and higher losses: a vicious circle [3].

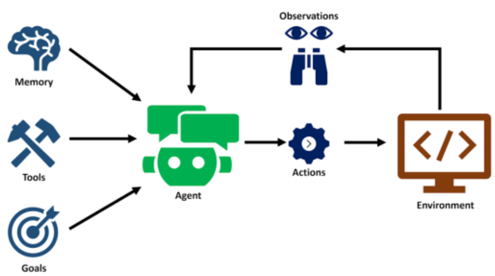

Fig 2: A deep dive into the future of automation

The wireless community is experimenting a change of paradigm. Solutions aided with machine learning (ML) are currently being developed with the effort of several groups and companies. Among ML techniques, RL has particularly received attention since it is an efficient approach capable of facing the issue of several states not being known a priori. Since RL uses trial and error, it learns through interactions with the environment and theoretically can cope with complex and dynamic environments. As alternative approaches, RL methods are still in their infancy. They need lots of interactions with agents and thus either are slow or not feasible for environment of larger order. Meanwhile, in most implementations, they rely on Q-learning algorithms and rules which are complex [6].

4. Agentic Intelligence Overview

According to [7], with the growth of the size and complexity of communication systems, it has become clear that the assumption of controlling and monitoring the systems through a central entity is no longer valid because of the limitations associated with a centralized approach for control. A central problem is to make the network self-knowledgeable, self diagnosing, and perhaps in the future self-managing with very little human intervention. Agent-based methods have been used in wide-ranging areas but their application in the cognitive spectrum management is a new and unexplored area. Intelligence, cooperation, learning, and adaptability are some of the important attributes of the conventional intelligence and this is now being investigated in the context of communications due to the demand of complex heterogeneous networks. In these architectures, the information and processing are distributed amongst the sensors along the cognitive network. Agent in the multi-agent system can communicate with the other agents, and the resulting behavior is a result of its observations, knowledge, and interactions with other agents. The multi-agent systems have proved to be an appropriate tool for automated system management due to their characteristics of decentralization and distribution, reactivity, proactivity, cooperation, autonomy and adaptation. Cognitive radio needs to sense and utilize the most suitable underutilized spectrums based on the service requirements. These operative decisions will be based on the hardware capabilities of the radio terminal like energy consumption, network type, transmission power, modulation, etc. Certain policies will be needed that will help us to provide the selection of a type of radio access networks compared to another. The association of MAS with cognitive radio can ensure a bright future for the optimal management of frequencies. In case of unlicensed bands usage, the cognitive radio terminals must coordinate and cooperate for better spectrum usage without interfering with each other. For licensed bands usage, the consideration varies with respect to time scale of channel quality, availability, licensing, etc. In this regard, cooperation is preferred to form a multi-hop network, federated networks or to become a relay.

4.1. Definition and Scope Nowadays, spectrum management has become a problem of extreme interest due to the spectrum's limited and precious resource. The absence of a centralized controller together with the artifacts of the wireless medium like interferences have made the problem very challenging to deal with. In recent years, some have proposed decentralized solutions based on game theory and defined an optimal or equilibrium solution to the spectrum management problem. But games are based on an idea of stationary strategies, while in reality, each user will have a dynamic strategy and thus a time varying decision of channel availability. Such time varying responses, that depend on the ambient data regarding the observed interferences levels, signal power readings, temporal nature of transmissions, etc., are complex and have motivated the work of many researchers to look into the timescale of the problem that naturally arises due to such increased complexity. One proposed a multi-timescale architecture and adjustment strategy for spectrum management, where the tuning of channel usage probabilities, the resource allocation strategy, and the on-off strategies were interleaved over different timescales. On a while, a biologically inspired Decentralized Adaptive Multi-timescale Budget Allocation Controller is proposed. Formulating the problem as a selection and allocation of budgets, their properties and guarantees as a solution via the proposed controller are derived. Biologically inspired ideas of population and selection of a behavior, and other methods used in year long experiments by biologists, like reinforcement learning, adaptation via simulated annealing, and pheromone retraction are invoked to construct novel non traditional and decentralized solutions to the problem of resource allocation of wireless communications [7].

Such approaches have been successfully adopted mostly in discrete spaces, while in uncountable continuous spaces the guarantees and properties of convergence are still unknown. Multi-Agent Systems (MAS) have also recently rekindled interest because of the multi-dimensionality of the problem, and a high degree of autonomy and diversity of models and behaviors that can induce new dynamics. In wireless communications, these models and theories can be readily adopted in conceiving decentralized, adaptive and robust systems that are capable of new emergent behaviors. On this line, an agent based MAS for decentralized, speed adaptive, cargo allocation in earth-moon transfer using pulse jet propulsion is proposed. Players with different communication bandwidth and capabilities coexist, holding different internal models to yield different emergent social models. These models interact in real-time to yield properties of goal dependent compression, criteria evolving multiplexing, and periodic explosions. In 2011, systems composed of multiple agents are presented. But, unlike proposed now-a-day cognitive systems, these agents do not learn or retain knowledge in the form of awareness about other agents or the environment or how their actions could influence it.

4.2. Comparative Analysis with Traditional AI Due to the rapid increase in wireless devices, the demand for the radio frequency spectrum has significantly increased. To safeguard communication stability, the spectrum should be orchestrated effectively. However, the existing spectrum management methods are personalized, closed, data agile, and complex. An AI-driven spectrum management platform is suggested to store data generated by various managing methods [8]. A competitive bid spectrum managing scheme is suggested, which guarantees performance similar to sophisticated decisions with low complexity. A reinforcement learning-based secondary spectrum management method considering spectrum auctioning is suggested. Furthermore, a comparison between suggested spectrum management schemes and traditional AI-based methods is discussed by analyzing computing complexity and communication cost. As typical AI creates the demand, deep learning and reinforcement learning are categorized as typical AI. Because of the model-less nature, communicating the trained model in addition to the data and the necessity of tuning every time for other dataset types incurs high costs in computation and transmission [9]. As spectrum management methods created demand, disclosed under well-known criteria; accuracy, time complexity, and communication cost of the methods, comparison with edge intelligence-based methods might seem redundant. Nevertheless, as spectrum management methods deal with emerging unseen demands, the aspect of fairness in an auction or resource allocation is a quality factor not affected in classical settings. Also, it must be noted that there usually exist well-known solutions for underground and visible domains to serve as benchmarks, while it is noteworthy that there aren't any in unseen or scattered domains. That is, baseline performance might be hard to be guaranteed, while exposed working conditions heavily affect performance. Afinig offers performance of previous spectrum management methods over other spectrums as a working condition. For clarity, domain uniqueness is also assumed hence in line with real applications, it's assumed that the algorithms trained privately or at least with needed inputs.

5. Dynamic Wireless Optimization Techniques

Dynamic spectrum management is a significant technique proposed for wireless spectrum usage optimization. This approach continually measures collapsible criteria such as service quality or users' demands and allocates wireless spectrum gains accordingly over time [8]. The resultant yield of high-quality wireless access can be converted to preventable costs or effective capital previously owned. Dynamic wireless optimization technologies aim to efficiently and successfully perform dynamic spectrum management approaches on a public basis. Such technologies perform ranging uses from on-demand allocation for dedicated users over limited resources with pre-designed schemes to neural-driven spectrum splitting across cognitive users with a success-driven course.

Two most-discussed methods for dynamic wireless optimization are elaborated on from their aspects of techniques, objectives, and performance in terms of efficiency and robustness. In many practical scenarios, including the fifth-generation system, the development of demand-aware dynamic spectrum allocation can supply unframed perspective on such usage. Typically, a channel settling to form a dynamic spectrum allocation system is chosen from all wireless spectrum resources at unseen cost using off-the-shelf algorithms, which has been easily and well replicated by industrial mobile operators. Unsolved earnest issues such as channel band effects on user dendrite and broaden coverage of spectrum allocation rage with high topology complexity remain in academia.

For large networks, a dynamic spectrum allocation metric is concerned for existing methods to eliminate usage of the inverse correlation measures of the channel reinforcement pattern and user reservation prediction. Using a mechanism as shadowing for several metrics is also highlighted in a few dignity assignment methods. Faster rates of traditional performance convergence and stronger robustness against delays and noise are shown. To a larger amount of channels pursued in views, on-demand dynamic spectrum allocation methods for dedicated channel assignment purposes are discussed and compared in solution space similar to the one-dimensional case with a closed property.

5.1. Frequency Allocation The frequency is reserved in accordance with the planned Spielberg coverage at altitude / suborbital height H by taking into account the line of sight from an earth station. The regulation and reserve of radio frequency spectrum has been suggested in , , as well as . Dubai will have to use an orbital trajectory tilted to the equator so that frequency reservation would have to be further planned in accordance with the earth stations on the trajectory [10]. This is particularly important when planned bi-directional multi-channels such as those in FIG must be efficiently used to reduce problems caused by echo interference and fading [11]. Therefore, especially with the proliferation of small satellites, frequency assignment must be considered and will be modeled first in . Would performance degradation be especially severe, if all the two banks are used unsynchronically? What would be the performance if both banks are used synchronically but the increment of delay is gradually increased? At first, as a preliminary case considering a pure case of eco suppression, all the two banks transmit to geostationary satellites and ground stations corresponding to those satellites respectively are employed anywhere on the globe. Furthermore, in order to estimate performance degradation, no other noise or interference is taken into account.

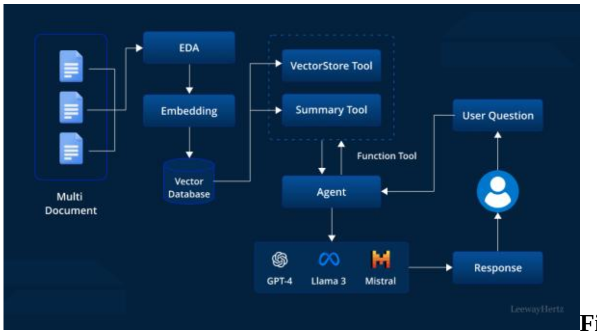

Fig 3: Applications and implementation

5.2. Interference Management With the demand for wireless communication systems growing, it is necessary to use the limited spectrum resource more efficiently. With the rapid increment of the user for advanced telecommunication services, it is believed that the conventional band allocation method is no longer able to satisfy the service requirements. Dynamic spectrum access can provide an answer to the problem as it allows the use of the unused spectrum band instead of collecting more bands. From the research perspective, there exists a lot of issues to improve the dynamic spectrum access model and performance. Specifically, there has been a considerable increase in the interest in intelligent spectrum management approaches that employ advanced control and optimization techniques including big data processing [8]. There is a need to analyze the current research trends and challenges in AI-driven spectrum management. It will provide the framework to formulate the research issues and directions for researchers in this area. This section analyzes the research trends and challenges focusing mainly on AI-driven spectrum management in intelligent wireless networks.

With the increasing demand for high data rate communication, the radio spectrum has become congested and cannot accommodate the exponential growth of demand. For efficient use of the limited scarce radio resources, the static spectrum allocation scheme will be assessed and the other means of allocating spectrum resources will be analyzed. AI-driven cognitive radio, called CR, can play a key role in spectrum management for next generation wireless communication systems [12]. With respect to CR, awareness is an important process that a user should know the state of the environment. An efficient and precise spectrum sensing scheme can help the user know the state of the environment better and this will contribute well to efficient spectrum management. Academic efforts to improve the perception capability of CR systems via advanced spectrum sensing methodologies will be analyzed.

6. Data Collection and Processing

Large amounts of heterogeneous and possibly time-sensitive information can pervade an entire wireless environment. In other words, on each channel at each observation time, the presence or absence of devices transmitting/receiving signals in compliance with one of several possible protocols needs to be predicted. Yet, with limited processing capabilities, particularly on boarding mobile devices that need to burn power and obtain results promptly, processing the perspectives of each channel would not only involve considerable processing but possibly also a delay in solutions, forfeiting the ability to engage in real-time negotiations with the user. In short, wireless spectrum data is often voluminous, heterogeneous, limited in processing and storage capabilities, and time-sensitive [13]. Regarding data acquisition, many devices such as spectrum monitors, spectrum analyzers, or protocol analyzers are deployed under the control of an administrator [8]. All of them communicate observational data to a central monitoring computer. Furthermore, a high-end cloud service provider executes the overall management framework. In this hierarchical layout, devices are subject to the operation of central management. Each of the spectrum monitoring devices produces and stores massive amounts of observational data.

A data processing design pattern suitable for the distributed architecture is envisioned. Data from various devices communicating the same data with the identical data format can experience the same processing and will be redundantly processed by divided devices in an asynchronous fashion. Conventional applications such as fraud and intrusion detection systems, with rules defined and needs of central control, have intended to deliver scaling up capability. This design pattern can also separate data-controlling curves, on-demand decision making, into either the devices and low-lying cloud or the ownership and high-end cloud according to the targets. Given a data format designed for a specific purpose and initial processing done on-device or on-the-edge, a line of self-organizing processes can convert the data from sensing to a pretty high-level form that can directly suit the needs of the mobile. Four data processing subprocesses are defined as follows: Data filtering for sampling and reduction, noise filtering for noise removal, the target fraction of channel modulations extraction for grabbing channel incumbents, and finally, modulation and transmission protocol classification for prediction about channels and devices rapidly corresponding to their possible measurements and signals.

$$\mathrm{SINR}_i = \frac{P_i G_{ii}}{\sum_{j \neq i} P_j G_{ji} + N_0}$$

Eqn.2:SINR (Signal-to-Interference-plus-Noise Ratio)

- P_i: power allocated to user i
- G_{ii}: channel gain from transmitter to receiver i
- G_{ji}: interference from user j to user i
- N_0: noise power

6.1. Sensor Networks As wireless communication networks proliferate, there is an ever-increasing need for radio spectrum. However, a large part of the radio spectrum is still underutilized, causing the fixed spectrum allocation policies to be ineffective. Dynamic Spectrum Access (DSA) aims to improve spectrum utilization by allowing secondary users to opportunistically access the licensed spectrum of primary users. The barrier of entry into Dynamic Spectrum Access is the need for spectrum sensing, which is the process of identifying how the spectrum is being used. The success of spectrum sensing relies heavily on the accuracy of the spectrum occupancy prediction algorithms. This thesis contributes to improving the accuracy of spectrum occupancy prediction in several contexts through machine learning. The first contribution is applying Artificial Neural Networks (ANN) to predict the spectrum occupancy of 802.11 devices. The results show that ANN is able to effectively model the output characteristics of the spectrum occupancy of the network's control channel. The second set of contributions consists of general architectures for spectrum occupancy analysis and prediction in which a collection of devices develop a joint learning strategy to predict the usage of their spectrum bands. These contributions present a multi-agent federated learning strategy and a method to optimally schedule concurrent learning rounds [4].

These devices collaborate in building a model, which they periodically share among them. This knowledge boosts the performance of local prediction architectures. A systematic method is proposed for devices to collaboratively schedule their concurrent learning rounds while considering device battery power states to maximize the learning rate. Formulations of the spectrum slicing problem from different perspectives, solutions, and approaches have been proposed and extensively studied. However, this area still lacks a framework covering the whole chain of operations. This contribution defines a formal framework that spans from the end-to-end slice design (dimensioning and instantiation) to the operational control and optimization, covering static and dynamic use cases. This contribution identifies gaps in the literature and outlines potential directions for addressing them [14]. Finally, applications and state-of-the-art results on DRL-based end-to-end slicing optimization problems in the context of 5G and Beyond are surveyed. The key challenges and open problems are discussed.

6.2. Data Preprocessing Techniques Radio spectrum management (RSM) has gained importance with growing demand for screen frequency band [8]. RSM is classified into static and dynamic types. Traditional RSM technology is static, but with the rapid development of wireless services expansion and RF equipment, fixed frequency allocation schemes no longer meet demand. Therefore, paradigms of spectrum management have changed, focusing on dynamic spectrum management (DSM) techniques. Spectrum mobility is to discover idle bands in the environment and utilize them on a dynamic basis. In order for spectrum mobility to be possible, new spectrum coordination techniques (especially central control architecture) worthy of an advance to traditional spectrum vacancy detection technology are essential. More advanced techniques are to detect spectrum vacancy with distributed control architecture, i.e., in a generalized fashion. To match the status of such demand, spectrum vacancy detection technology is required. This technology is also fundamental to any wireless service operating in fcc-licensed bands. Numerous approaches to spectrum vacancy detection are already alluded to in academia. However, there is no overall comparison to clarify the state-of-the-art methods. In this context, passive spectrum-vacancy detection is firstly addressed, and a general classification of existing detection algorithms is presented upon summary of detailed contents on each scheme.

In addition, strengths and weaknesses of sensors are described, along with rigorous performance evaluations via extensive simulations. The second important aspect is to show practical issues involved in the real application of such algorithms. In this context, topics like multi-channel processing, the longer temporal correlation of band occupancy state, and wide-band spectrum, are explored [15]. Cognitive radio (CR) is the main implementation platform. In order to enable hazardous wireless service operation in a licensed band (L-band), spectrum mobility is to be incorporated in can bus-based CR nodes system with a goal of narrowly understanding both theoretical and practical research interests in this area. Of the myriad of necessary components, specifically spectrum sensing and frequency-spring time-offset staring are focused. The application of modified DIDCM to can buses with a common time reference and extensive simulations show that the proposed methods can successfully enable adaptive and vehicle-to-vehicle communications in a mighty manner under both time-varying and static traffic situations.

7. Model Development

Spectrum Management (SM) is a pivotal process to guarantee the performance of communication systems in dynamic environments. SM includes the sub-processes of Spectrum Sensing (SS), Spectrum Decision (SD), and Spectrum Sharing (SH) [5]. With the advancement of connectivity technologies, the demand for spectrum resources has rapidly increased. However, governments around the world presume spectrum needs for CBR designs to assign exclusive licenses to communication operators. This leads to excess demand for area-wide accessibility to spectrum resources and is a critical bottleneck for new connectivity systems. To efficiently utilize transiently under-used licensed spectrum resources, researchers have been investigating Bandwidth Reuse Technologies (BRTs) which enable WidowDists/Bands (WDs/WBs) to sense and negotiate participating in exclusive licensed bands and bandwidth in a dynamic manner.

Considerable efforts have been devoted to the theoretical investigation of SM sub-processes. With the serial and manual structure of SM processes, systems can merely allocate a single WB to one WDs via heuristic detectors utilizing spectral-spatial statistical models. However, in real systems, SM is more interactive and simultaneous due to dispersed channel conditions and non-stationary environments. WD also would like to negotiate and use WBs from multiple bands simultaneously in order to provide multi-channel service. Additionally, the increasing bandwidth demands of successive communication systems also stimulate the usage of wider bands, leading to the rapid development and deployment of WBs of hundreds of megahertz that have been assigned as exclusive licensed wide bands for MNOs to promise QoS and eliminate interferences.

Efficiently managing under-utilized wide bands for ultra wireless systems has become a new research focus. However, the non-stationary properties of spectral-spatial patterns and interferences exhibit intricate challenges for selected resource allocation in the wide band. Also, the limited spectral legacy necessitates the collaborative management of the visible spectral bands and their wide integrated spectrum resources in a dynamic manner. Since WDs/WBs and wide under-utilized bands' availability are dispersed, non-stationary, and largely invisible, Bandwidth Discriminative Technologies (BDTs) are required to capture detailed spectral property dynamics for device-wide under-utilized wide band selection. Due to the inconsistency of historical data and the illegibility of spatiotemporal features, existing supervised modelling of BDTs would not be suitable.

7.1. Feature Selection In this section, common feature selection methods for improving the classification process are classified into four types, and methods are designed to perform search-based feature ranking. If the input training data set is correctly selected, machine learning algorithms may considerably increase spectrum sensing performance. The benefits and drawbacks of using input data sets with energy detection assessments and energy values in the actual world were investigated by [11], who found that either input could provide reliable results over longer time period periods, but that both input datasets were impractical for real-time applications. [16] also researched inputs with energy values. Low-complexity algorithms for increasing the number of users while using the least amount of power, and methods for decreasing users while using the most amount of power, were developed to solve user selection with power allocation. The neural network was able to rapidly learn the best user selection strategy in an unknown dynamic environment with a high success rate and fast convergence.

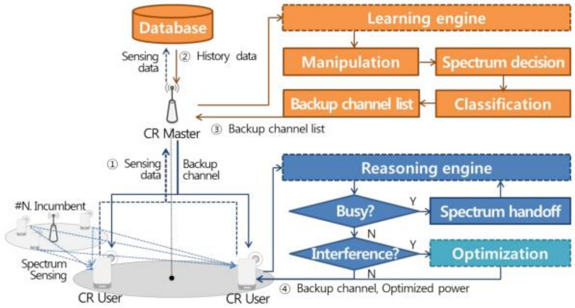

Fig 4: Intelligent Dynamic Spectrum

The inputs of power detection methods are energy detection and thresholds. Decisions were made on this input and the effects of interference and waste were evaluated for a range of occupancy rates. Using fast data input that includes power spectral density estimation technique which is well known and widely used in time frequency analysis, spectrum sensing analysis by using classification algorithms were evaluated regarding trade-off between the detection accuracy and analysis time length. Anomaly detection methodology using time pattern analysis was studied, which extracts time to detect abnormal period when high usage is encountered. Probabilistic model to detect idle state for CRs was introduced considering for a given level of false alarm. Reinforcement learning and policy-based multi-agent systems were suggested. The suggested method adapts Q-learning and PCA (for user channel selection) and light-weight SVM to schedule requests. The approach shows that the nodes are able to make autonomous choices about which channel to use and how to move between multiple channels.

7.2. Model Training and Validation This section presents the specifications for the data generation and setup used in the network simulator. The developed OPNET simulation model is discussed, and each critical block is described in detail, including its function and implementation. 5G New Radio (NR) and 5G-NR offset-time generation processes that assist the spectrum contention process that substantiates the cognitive radio environment are also described.

The simulation was carried out using MATLAB R2020b, which was chosen because it has many different toolboxes available, which could be used to design the entire system. As stated previously, the framework consists of an OPNET simulation model that runs the 5G NR, LTE, and LTE-A application layer, a MATLAB toolbox that runs the SLRMA algorithm, and a graphical user interface that displays the simulation results, such as monthly graphs. The application layer of OPNET was used to generate fives scenarios containing thousands of source nodes. It runs simulations that get frames ready to be sent into the queue of the RF channel. It includes pre- and post-routing stages when frames are converted from L2 down to L1 packets. This simulation model was developed using OPNET, which can be expensive and difficult to work with.

Each of the simulation scenarios ran for 1 minute, with the first 10 seconds used for warm-up. During this period, the necessary applications, devices, and network parameters are analyzed by the application layer and fixed receiver parameters and offsets are set. After the load-stressing period, there is a controlled degradation where nodes try to change their channel and parameters every 2 s. Results are presented in different perspectives that describe the evolution of the simulated environment and its behavior over time [4]. Some are basic metrics, such as the number of satisfied packets generated and accepted, while others are more sophisticated, such as the cumulative cost of failures in terms of wasted bandwidth and required retransmissions. The complete observation of the simulation can be seen in the accompanying numerical and graphical summaries.

In the simulation framework, each base station runs a brain that controls its domain devices, states, behavioral policies, and parameters. They communicate with the distributed monitoring framework to share this information with the other base stations. All nodes maintain a local database of their calculated information that is updated frequently and assists in predicting spectral occupancy [5].

8. Implementation of AI Models

Spectrum management (SM) is an important aspect of wireless telecommunications and therefore needs to be automatic and precise over time, user and geography. An ultra-reliable low latency communication (URLLC) intelligent radio network (URN) can be defined of sensor, software-defined and cognitive radio networks able to sense and autonomously manage spectrum resources without human intervention [5]. The radio resources management (RRM) under pinning functions should be automated through AI with, among others, SM a wide belt encompassing a set of functional domains. A classification of techniques enabling AI-powered SM RRM is proposed, discussing about advantages and drawbacks with respect to the specifics of wireless communications. The need for this type of access arises from the saturation of the frequency bands both in the government and commercial spectrum with many devices that operate over them. The frequencies are divided into channels (used by one or many users) and spectrum holes (unoccupied portions of the spectrum). Both users and channels have characteristics that change over time and stochastic modelling is needed in order to adapt to their variations. A semi-competitive intelligent agent settle into a decentralized self-organized cognitive radio scheme is described too. Its behaviour is described as a multi-scenario multi-user single-channel one where all the users try and access the channel on a first-come-first-serve basis. To find the right time to transmit, CRs need to have some basic senses on the spectral environment. They may go through a trained neural network or some other kind of time-consuming means in order to obtain this information. Wideband environments are more difficult to monitor than narrowband ones, so the framework is aimed at providing algorithms that make this task as less expensive as possible.

8.1. Deployment Strategies End users and service providers commonly utilize spectral resources in a heterogeneous method. As new technologies and services are introduced, more spectrum bands become available. Service Providers are therefore motivated to maximize their monetary profits by allocating provided spectrum resources in the most capable manner. Therefore, a provision for efficient spectrum utilization, channel allocation and handoff management under the imposed constraints of a spectrum auction, is an efficient and comprehensive provision of RF spectrum management.

With the growing focus on intelligent cognitive wireless technologies and additional frequency bands being offered by regulatory bodies, there is a demand for more operating frequency bands available for Wi-Fi. Typically, a fixed number of channel frequencies are accessible where access point options are always unchanging. But if there are any underutilized channels, users and devices cannot switch to another non-used band autonomously without shutting down and switching. Service-wise, a further demographic of client requests is on the rise, whereas bandwidth-hogging apps for streaming media are restricting spectral utility. Introducing machine learning to cognitive radio technology can dynamically allocate unused channels in real-time to a configurable network of devices, such extending a Wi-Fi coverage while further band plans and privacy policies can be preserved [11].

A new system levels the channel allocations in parallel but separates the cognitive and regulative resources on differing management layers. The autonomy of the allocation system, and agility of AI agents' communication is simulated as Internet Protocol over the physical layer. Experiments show the efficacy of cascading the channel finder's and the channel selector's learning by adding watermarks to a mutually experienced channel state list and policy completion time testing which can reproduce the falsification of an individual channel's probability detection metrics when modulating the cognitive knowledge base's state space [8]. This study will enable the improvement of not needing to restart learning after the individual network configuration changes through some variable knowledge base checking for any incomplete epochs.

8.2. Real-Time Processing The basis for AI-driven spectrum management is real-time processing. In real-time processing, the time duration for processing data is constant, so adequate amounts of computing resources should be allocated to application workloads for processing. If the computing resource is insufficient, data may be missed or delayed, which cannot be tolerated in real-time applications, such as industrial Internet of Things (IoT) and tactile Internet [8]. There have been a variety of strategies for real-time data processing, and a major characteristic of real-time processing compared to traditional big data processing is that there are strict bounds on the time accepted for data (or events) to be processed. Real-time processing can be defined as that all processing workloads for input data should be completed before their deadlines. Time, input, and output are rarely continuous; instead, they tend to be consumed one by one in the order of arrival for transiently stream data. For the alternative input or output flows, the corresponding entities of the other types can be generated at each event arrival that conforms to certain rules. Processing the event caused by the arrival of an input entity basically consists of one-to-many message delivery. Only when all deliveries have been completed does the event become a one-to-many output entity [13]. The real-time processing should guarantee that a given workload of pre-distributed streaming queries completes before its specified deadlines. A major necessity to enforce this condition is that resource allocation should be performed in real time. In the machine learning (ML) model-based brokerage of spectrum resources, several ML models are trained using learning datasets in pre-processing and form a broker to allocate spectrum resources to IoT devices in a real-time manner. Specially for an emerging query of an IoT device, the broker is composed of two processes: inputting a query and processing it. Given the query, features of its properties and model processing are extracted as input data. The features are fed into the broker and then processed by the trained models.

9. Case Studies

This section discusses two case studies of AI-Driven Spectrum Management in Wideband Multi-UAV Traffic Management Systems by [5] and Intelligent Dynamic Real-Time Spectrum Resource Management for Industrial IoT in Edge Computing by [8].

Multidrone networks for UAS could emerge as a solution for a wide range of applications in different industries over urban environments. Whereas industrial, scientific, and medical (ISM) bands are standard for UAV communication, this spectrum is already quite congested. The US Federal communication commission (FCC) numbers to more spectrum access options towards the success of future UAS. However, prior to doing so, it is of utmost importance to develop efficient spectrum sharing policies among the existing drone UAVs to enable compliance with regulations while ensuring public safety and quality of service. Moreover, wideband spectrum management service to sense, assign, and monitor spectrum usage across the entire frequency band is an important consideration for effectively enabling advanced UAV use cases. In this paper, they propose a data-driven model for wideband spectrum sensing and scheduling across multi-drone UAVs acting as secondary users to opportunistically utilize detected spectrum holes. In this architecture, state-action pairs of the proposed-wideband spectrum scheduler with multi-UAV can be modeled as a Markov decision process (MDP). To this end, the energy efficiency in the perspective of secondary publishers with cooperative sensing, coordination, action searching, and scheduling across the multi-drone network is devised as the metric to solve the proposed MDP-based problem.

The basic task is to classify a set of observed I/Q samples based on whether they belong to spectrum holes or not. The proposed multi-class classification framework is constructed as a fully connected feed-forward network and will leverage case studies with comparable metrics and design specifications to train the auto-encoder network. The objective of a classifier within a case study is to label observations generated by that case study on detecting spectrum holes. They will adapt it to newly occurring case studies with refinement and fusing to probe capability and robustness in diverse spectrum usage patterns.

Deep Q-Network (DQN), a pioneering deep reinforcement learning (DRL) algorithm with great performance in learning elegant policies in unstructured and high-dimensional supply problems, is first considered as a benchmark in this chapter. To mitigate overestimations in Q-value predictions from a DQN agent, double DQN (DDQN), with separate target networks for action selections and target Q-value predictions with the same parameters further explored for freezer techniques, replaces the standard fixed target structure in vanilla Q-learning. To promote learning stability, DDQN agents are examined with soft updates for the action-value network and target network coupling with Q-value uniformization.

9.1. Urban Environments It is important to realize that there are various spatial configurations of urban environments, all of which have impacts on how radio signals propagate, thus affecting how radio spectrum resources should be managed. In some cases, buildings are distributed evenly along the streets whereas trees grow abundantly along the sidewalks; while in other cases, there are canyons formed by tall buildings on two sides, with flat lands around them. Also, arch bridges seen in many cities can block radio signals. As such, it is necessary to understand the characteristics of the environment at the beginning of AI modeling as they can be significant inputs to the modeling tasks. Those characteristics can be categorized into two levels: properties of the whole area and local properties of each point in the area.

For urban environments, properties of the area can usually be represented with a density map. However, conventional density maps of buildings are too rigid to capture real-deployed irregular situations, such as jagged edge and concave corners. In this regard, a novel, parameterized representation is proposed for urban buildings that involve two levels of input data and an equivariant value function. The former include discretely-computed spatial regions for objects and their geometry parameters, while the latter consist of extracted shape descriptors for this region in addition to the geometry parameters. By designing a neural network to estimate values as a function of both the shape descriptors and geometry parameters, the predicted values would intrinsically be equivariant with respect to the geometry parameters and robust to noise in the shape descriptor extraction. Based on these ideas, experiments are provided to evaluate the performance of the design of density maps along with potential applications [1].

For local properties of each point, urban environments can be represented as a graph, in which the vertices represent the signals, environment, and resources, while the edges indicate the relationships and interactions between those factors. In this representation, speed and energy of the signal are considered velocity of edges, and if a vertex is reliable is determined by its connected edges. Existing approaches utilize DQN or A3C whose training efficiency is sought. In this regard, a multi-task RL method is proposed to simultaneously train the policies for the adjustment of the receiver noise, the transmit power of the uni-input uni-output (UIO) and multiple-input single-output (MISO) channels, and the operator selection for the non-orthogonal multiple access (NOMA) transmission scheme [4].

9.2. Rural Deployments Despite significant growth in communication infrastructure, 52% of the world’s population remains unconnected, mostly in developing countries [17]. The access to broadband is worse for this poor rural population due to the high cost of infrastructure, difficult terrain, sparse population density, and low Average Revenue per User (ARPU). Recently, the unconnected need to be connected for development has been realized worldwide. Low-cost broadband access can be provided by deploying Wi-Fi Access Points (APs). However, nearly all current technologies require laying fiber to backhaul each Wi-Fi AP, which is infeasible due to time and cost. If an optical Point of Presence (PoP) is available nearby, a wireless connection can be established to connect the PoP to the Wi-Fi APs. In many developing countries, the TV UHF band is highly underutilized and is a possible candidate for this middle mile network.

The TV band is highly underutilized in many developing countries, also available due to interference-free communication opportunities between white spaces and TV bands. In India, more than 100 MHz of this band goes unused due to different requirements for geographical regions, including size and parameters of the TV tower. The propagation characteristics of this band are favorable: it is possible to obtain a large coverage area even with low power transmission. This enables the use of renewable energy sources to develop technology that is affordable and sustainable in rural areas. Nevertheless, a middle mile network using the TV UHF band is not a plug and play solution: this band can be reused for communication by multiple operators. As a result, these operators can cause interference among themselves, a critical issue in Wireless Networks. This low availability of spectrum resources leads to low spectral efficiency and a large number of users being unable to connect. The agenda is to design a spectrum sharing scheme for the middle mile network for the rural setting that increases spectral efficiency and guarantees a fair share of spectrum to all operators. In any rural area, however, the availability of any prior information is unlikely due to the deployment of newly developed networks, wherein such information is often not explicitly available. Because of this, the operators are often unwilling to share sensitive information that could lead to an advantage with respect to the competition. Hence, the challenge is to design a scheme based on easily available information [18].

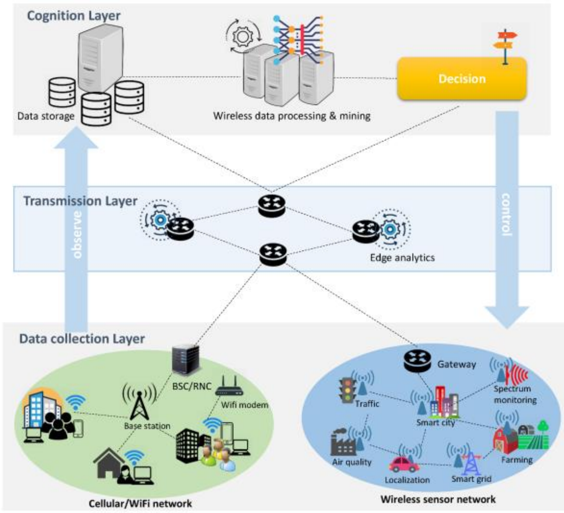

Fig 5: Based Performance Improvement of Wireless Networks

10. Challenges in Spectrum Management

There are several challenges in the spectrum management process, tools/technologies to be developed to overcome these challenges. There always exists a demand of spectral authorities to be able to characterize the spectrum usage, mainly to ensure the continuing relevance and viability of regimes like the fixed and dynamic bands, which are key to many wireless communication systems. However, the provision of energy or time monitoring services to a wide range of third parties, is a careful balancing act since a lot of this information may also be useful to potential interference causing entities [7].

dataset-driven analysis of spectral usage, allowing easy comparisons of spectral activity across different geographical regions, licensing regimes and spectral bands. heighten the awareness and understanding of spectrum sharing opportunities by nondomestic entities, assist engineers in the design of robust transmission schemes which are resistant to system performance degradation when implementing new devices, standards or schemes. Exist methods, exploring both computing mobility techniques and physical verification methods, for generating accurate and extensible spectral black-box models to grant a third-party access of communication devices under a behavioural economy, to be further investigated. Specifying what detection of third-party devices may be provided to its spectral authority could also be an interesting line of original research. Address the limitations of the existing analysis and prediction techniques both from an analytical and modeling point of view. Address a number of key challenges in intelligent wireless communication systems in general. In particular, examine different aspects of enaction-driven spectrum management, which is intended to automate the way that the energy vectors of available frequency channels are acquired, understood and used by computing engines. Also examine questions like how to better understand the existing knowledge of communication schemes, how to reuse them in a new geographical or spectral region, how to predict and understand what schemes are likely to be introduced next and what communication devices/standards may benefit from a detailed analysis of already active schemes.

10.1. Regulatory Issues The utilization of Artificial Intelligence (AI) to support many objectives will be immensely powerful in improving regulatory processes and procedures. Spectrum regulators are being faced with a number of challenges. There is a strong need to regularly refresh and revise regulatory frameworks, procedures, and processes. Common issues to regulators include how to deal with new technologies: primary and secondary spectrum allocation; market-based incentives; inter-technology and inter-service interference/pollution management; international harmonisation/alignments; and asymmetric national interests [19].

Today, many of these objectives are being addressed manually, with some reliance on ad hoc software applications. The demands for more agile, timely, and nuanced responses to these complex, long-term objectives far exceed the human capacity to manage them. On the other hand, the rapid maturity and diffusion of AI technologies are opening paradismantic possibilities and opportunities. AI capabilities in understanding natural language text and generating nuanced but meaningful responses have reached new highs. Massively scalable and cloud-native foundational Large Language

Models (LLMs) form the basis and the backbones of an AI-first approach to enhancing spectrum regulatory processes and addressing the highlighted objectives in a collaborative, semantic way rather than with hard coding based modelling.

Over the last two years, a number of Large Language Models (LLMs) have been made available to the public in various forms. The two most well-known LLMs are GPT-3.5 and GPT-4. By and large, these LLMs are based on the transformer architecture, which is a type of neural network architecture. These LLMs apply widely used natural language processing techniques, such as self-attention, those with which the LLMs have been trained on massive amounts of data [20]. These LLMs are designed to generate text in natural language in response to prompts. Other developers of LLMs include Google, Anthropic, Meta, Cohere, Mistral, and Hugging Face.

10.2. Technical Limitations Although AI-driven spectrum management has a great potential for improving system performance and increasing quality of service in tomorrow's wireless communications, the major challenge is that of obtaining reliable data so that the AI techniques can be trained and learn patterns that allow the prediction of the behavioral characteristics of the systems. In many practical situations, the systems are dynamic and the environments change [8]. Therefore AI-driven techniques should be studied against evolving environments providing challenge dataset scenarios.

Also, another limitation is that AI-driven spectrum management will likely be implemented in situations where precision is needed and the cost of the incorrect decision has a significant weight. For that reason, a fair assessment of the benefits that AI-driven techniques may bring should be evaluated based on specific applications, required precision, operating conditions, and potential risks and penalties. The controlled risk could be included as a penalty on the final performance measure, usually addressed by such techniques.

$$R_{global} = \sum_{i=1}^{N} (\beta \cdot \text{Throughput}_i - \delta \cdot \text{Interference}_i)$$

Eqn.3:Multi-Agent Coordination Reward

- Agents are trained to maximize global reward through collaborative or competitive learning.
- β, δ: tuning weights for balancing throughput vs. interference

Another item to be dealt with is that of input data description and consistency. The definitions of input parameters such as power availability or RF environment description should be done intelligently, without significant misuse of the resources. Nominal conversions should be performed cautiously so previous quantitative data can be retained and compared. The deployment of learning techniques over simulation needs to assure that the training dataset used for fitting the model is representative enough of the environments and scenarios in which the techniques will be finally operational.

Finally, it is important to remark that AI-driven approaches will likely not completely replace current methodologies. Most likely, such approaches will augment existing techniques in a way similar to that of the decisions systems that aid system operators. These systems commonly highlight the path decisions, inputs that carry risk and warrant caution, or statistics that help predict the quality of the decisions.

11. Future Trends in AI and Wireless Communication

Today's wireless communications and the Internet face enormous hurdles to satisfy the extreme data rates specified by today's and future wireless applications due to an exponential growth of data. Furthermore, these requirements are anticipated to grow exponentially in the coming years as a result of new applications such as Remote UAV (Unmanned Aerial Vehicle) Operations, Self-Driving Cars, and AR (Augmented Reality) & VR (Virtual Reality). For example, for Self-Driving Vehicles to operate safely, data of several Giga Bytes per second must be delivered with transport delays in the ms (milli-seconds) range. This data must be delivered Reliably with Minuscule Packet Loss. Due to the abundance of devices and the corresponding data created, the Internet of Things (IoT) is another method that will soon be generalized to billions of smart sensors. Soaring demands on storage, processing, and communication networks are anticipated as a result of all these devices producing data that requires severe filtration at a very early stage. Furthermore, responses to these decisions must be rapidly dispatched through the network. Therefore Ultra-Reliable Low-Latency Communication (uRLLC), Massive Machine-Type Communication (mMTC), and Broadband (eMBB) data flows must be supported for these devices. Unfortunately, there will not be enough spectrum in the forthcoming millimeterWave (mmW) or Terahertz (THz) bands to kind any feasible solution unless a radical shift in wireless network architectures and operating paradigms is executed.

Revolutionary advances in AI (Artificial Intelligence), ML (Machine Learning), and DNN (Deep Neural Networks) may offer a natural means to deal with the enormous data overload anticipated in tomorrow's wireless networks. The principle of AI that increases the QA (Quality Assurance) an order of magnitude by managing the entire network from end-to-end with minimal human intervention will be embraced. A data-driven global approach to Network Management will be adopted for the first time based on ANNs (Artificial Neural Networks) where interactions among the network elements will be learned from the recent network States (the history of all significant events occurring in the network). Reward Signals with primitives such as Adjacency, State Accuracy, Deployment, and Environment Awareness defined. The Traffic Given using SDGs (Spatial Decomposed Graphs) which capture all important features of a wired or wireless network: Nodes, Links and the Attributes (Service, Capacity, Cost, Latency) State Graphs & Request Graphs will be learning. Friendly interfaces to 3rd Party (domain experts) will design New Applications or New Domains. So in a few seconds, the Network will be massively reconfigured through flow alterations, routing changes, topology adotments, and hardware up-scalage [21].

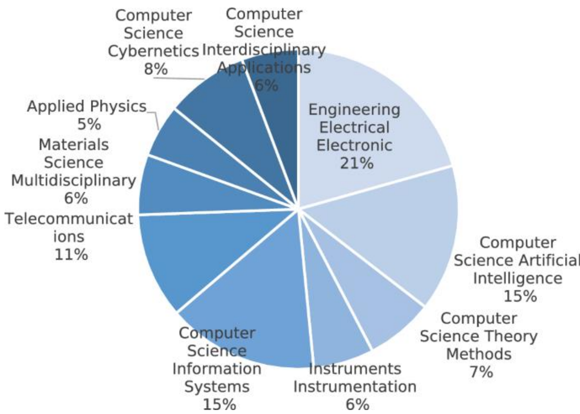

Fig : Artificial intelligence powered Metaverse

11.1. 5G and Beyond Since the launch of 1G networks more than thirty-five years ago, cellular technologies have undergone a number of evolutions. Each evolution, referred to as a "generation," was typically marked by a major set of new services and required advancements in the underlying technology, such as the air interface. Recognized by standards organizations and trade associations, these generations are commonly referred to as 2G, 3G, 4G/LTE, and the emerging 5G [22]. Standardization processes in third generation partnership project (3GPP) are ongoing to define the next generation of cellular technologies. Current timelines indicate that 5G deployment could commence as early as 2020. An extremely heterogeneous environment consisting of many different types of radio access networks (RANs) will present a formidable challenge in the management of the Service Providers' (SP) networks. The basic building block of such a technology-agnostic NSI is the realization of the concept of Software Defined Network (SDN) and Network Function Virtualization (NFV). To attain this, the development of Network automation capabilities is inevitable. These capabilities can be developed with the help of 'Artificial Intelligence' (AI). Mobile networks, which have been to date fairly manageable with SLAs and threshold based alerts, will evolve to networks which would be beyond human comprehension both in terms of scale and complexity [21]. The advent of 5G with its ultra-low-latency requirements will further complicate the overall management of these networks. Traditional means of management and measurement where the collected KPIs from the network would be processed and acted upon are expected to be insufficient. Hence, more automated solutions to different aspects of network management are required. One area of future work could be the design of a machine learning technique focusing on disaster recovery which is a key area of differentiation by operators. Given its complexity disaster recovery mainly relies on human expertise and experience. Focus here will be the collection and processing of AI data in order to provide the network with a state of preparedness prior to the occurrence of a disaster. This AI data will be in the form of collected KPIs from active measurements and modelled by clustering and cross validation and provided to the self-healing agents on the controllers. Each simulated state could serve as a disaster scenario with which the self-healing agents can be subjected to reinforcement learning.

11.2. Integration with IoT The rapid advancement of Industrial IoT (IIoT) has been accompanied by a tremendous increase in spectrum resource demand for connection among devices. It is necessary to have intelligent and dynamic real-time spectrum resource management capable of rapid and guaranteed handoff operations using cognitive radio technology. However, the existing intelligent spectrum resource management depends on the framework of consecutive reconsideration, which cannot sufficiently consider time-variant environments and relationships among devices. Hence, it is difficult to generally apply the existing intelligent spectrum resource management with convex optimization, a consideration that many devices may need to handle. To address these issues, it is proposed a novel intelligent dynamic spectrum resource management method based on reinforcement learning. The proposed resource management can use various learning models considering the relationship among devices [8].

For the defined goal and use conditions of each radio device, the Q-table is established with each MDP state, action (CSCH), and cumulative monthly reward value (Q-value). The optimal MDP policy of every radio device is dynamically generated by Q-Learning algorithm processes every time long-term CSCH recognition requests are received. For systems with a large number of devices, each radio device that does not review CSCH need can be generated with lower dimensional MDPs based on the consideration of the burst state of various use conditions. The proposed method can enhance connectivity and structural efficiency by providing a large number of spectrum usage combinational learning for a long burst of time [23].

The impact of the existing spectrum resource management is analyzed with performance evaluation. The settings and definitions describing the environment are illustrated in Section 5. Section 6 introduces a series of tests with respect to the defined use conditions, and details on the design process of RL-based spectrum resource management are addressed. The existing resource management is tested under the same conditions for performance comparison, and demonstration diagrams illustrating the proposed spectrum management are shown. Finally, a conclusion is proposed.

12. Ethical Considerations

An examination of ethics surrounding AI manifests, regardless of its application or implementation. These considerations can vary dramatically based on the intent of the engineers and scientists involved in the design, development, and implementation of AI applications. Engineers and policy actors may grapple with social implications or the potential effects on human lives, jobs, security, and safety. Business executives and lawyers may contemplate public relations goals in light of reputational risk and potential liability. This discussion focuses on whether AI applications that subvert or undermine obligations that do not promote, protect, and respect fundamental human rights can be characterized as ethical [24]. In terms of application and impact, AI is not unlike pharmaceuticals. It warrants consideration as a category of products with the potential for both good and harm, rendering it subject to the appropriate level of regulation.

The relevance of AI to human rights is determined by its diversity of manifest forms. Where the AI is computing in some impactful way, its potential harms must be recognized and minimized, and it warrants stipulation of rights protection. Where there is reliance on statistical practices, AI applicability is, by definition, relevant to ethics, as many AI applications will be some statistical practice, often seeking to draw inference or make prediction from data [11]. The process of AI design, development, deployment, or utilization should include Human Rights Impact Assessments, prepared by qualified individuals or teams independent of the builders of the AI, before it is considered ethical to proceed. Designers, developers, deployers, and users of AI products should consider whether they are impacting fundamental human rights.

As simply pointed out, “AI ethics or fairness, accountability, and transparency (FAT) in algorithmic designs and applications involves a myriad of legal and moral questions and issues.” Given the growing deployment and opaqueness of AI in design, production, distribution, sale, and use of goods and services, conformance with product and market regulations may be insufficient. The stronger expectation of design and application fairness, accountability, and transparency raises challenges for able observance. Challenges with observance include corporate cultures resistant to disclosure and the internet infrastructure indexing, interconnecting, and perpetuating in a complex web vast numbers of AI-based applications, often uncontrollable by their designers and builders.

12.1. Data Privacy

Demand for wireless bandwidth is rising much faster than new spectrum assets can be made available. Thus traditional command and control spectrum management is threatened with collapse [25]. Ad hoc solutions based on secondary access could enable the sharing of the unutilized spectrum. Cognitive radio (CR) addresses these solutions using reconfigurable radios for opportunistic access based on spectrum sensing technologies and elaborate optimization theories. However today SM is performed statically based on technology and policy. Users can tune their transmitters to some pre-allocated channels based on regulations. They might be appliances of fixed services or base stations of cellular communication systems. New concepts to solve these looming spectrum problems include dynamic spectrum access (DSA) or open spectrum access. Through primary access to unallocated frequencies could significantly expand the resource pool for CRs. In contrast, secondary access focuses on the use of bands exclusively licensed to other users for free.

Next, the resource analysis will highlight recent attempts to develop intelligent spectrum management platforms using a combination of sensor networks and sophisticated algorithms. Such software-based approaches to optimize the use of existing band resources will be explored. Energies could also be harvested from ambient radiation which could be exploited for opportunistic communications. Techniques to assist ultra-low power and ultra-low bandwidth communications through opportunistic wake-up receivers will also be examined. Further, integration efforts to fuse wider band resources for each communication will be explained. The best example to showcase the value addition of spectrum resource management would be the Global Navigation Satellite Systems. Ways to organize overlapping offering capabilities and exploit more satellite resources to enhance the accuracy and robustness of navigation services would be explored.

12.2. Bias in AI Algorithms

Artificial Intelligence (AI) is having a major impact on our lives. From helping us book flights or hotels to driving cars or diagnosing diseases, AI technologies are growing in precision and complexity. However, as these technologies are increasingly trusted with performing tasks for human benefit, concern is also being raised over unintended mistakes made by these systems that have the potential to cause massive harm, be it to our resources, privacy, or even in terms of human lives lost. Unfortunately, theories proposed by the scientific community to understand, trust, and validate these systems aren’t keeping pace with technology. Among the biggest challenges and sources of controversy around AI are questions of bias. The literature surrounding trust in AI-based technologies is largely silent on the topic of bias. Although algorithms themselves are technically proficient, bias inherent in those systems can manifest through data, assumptions, processing methods, and sampling, resulting in algorithmic bias. Ideally, the entire design of an AI system is fair. Yet reality demonstrates that algorithms are only as fair as the datasets are inclusive, and researchers are currently tackling the problem of ensuring data fairness. However, bias can exist even if the data is fair, and the learning algorithm can exploit that bias, which results in a set of outcomes that violates the principle of equity. Therefore, understanding the biases inherent in the algorithms becomes paramount [26].

Bias is an algorithmic and systematic error in judgement. For example, in the context of loan applications, algorithmic bias may mean that an applicant's sex, age, or ethnic made predictions more likely to lead to loan denials. Human reverse perception may help identify biases, but it is only partially able to identify algorithmic bias in high-dimensional data due to the complexity involved. There are two types of algorithmic bias: inductive bias, which is inherent to the design of the algorithm itself, and specific bias, which is encoding a specific characteristic of the input training set. Although bias-free algorithms exist, the prevalent concern is the presence of bias in AI-based systems and many response techniques exist for its mitigation, including altering data preprocessing techniques, altering algorithmic design, or political interventions [27].

13. Conclusion

Spectrum resource management is of paramount importance in multi-access edge computing (MEC)-enabled vehicular networks with multi-service applications. This paper presents a spectrum resource management scheme with the aim of maximizing the network utility by optimizing the BS-user association patterns and spectrum slicing ratios. The proposed scheme, namely the Spectrum Resource Management Framework (SRMF), is to maximize the summation of the achievable transmission rates of individual AVs from BSs, while the resource block-user association patterns of each tier are recursively updated. One-tier spectrum slicing at the MEC edge with guaranteed channel access and the calculation of the summation of the achievable transmission rates of AVs from BSs are designed based on the newly fixed user association patterns. Extensive simulation results show that the proposed scheme achieves better performance compared with some benchmark schemes. Spectrum resource management for multiple access edge computing is of vital significance to accommodate the emerging devices demanding low-latency service with various applications in modern vehicular networks [1]. However, the increasing number of AVs with stringent ultra-reliable low-latency communication (URLLC) requirements poses a significant challenge for spectrum resource allocation, especially when considering channel access and high dimensionality.

The traditional frequency division doubling or frequency division multiplexing offers limited capacity gain. It is critical to develop low-cost, high-efficiency, and high-speed spectrum resource management mechanisms with traditional and unlicensed techniques adaptable to a dynamic transmission environment. To examine the possibility of characterizing the underlying regular spectral structures, a few initial features are captured, including the frequency bandwidth and channel lookup duration as state variables and the assigned channel center frequencies and bandwidths as control variables. However, the features are domain-specific and task-specific. The dynamic and complicated pollution source management across the frequency domain hinders extracting a group of generic features that can be adapted to other scenarios or platforms. The continuous improvement of the quality of the first-feature-extracting method can effectively reduce the upper bound on temporal dimension and improve modeling performance [8].

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